13.10.2026

(P1) $\frac{\partial^2 f}{\partial x_1 \partial x} = \frac{\partial^2 f}{\partial x_2 \partial x_1}$

Order doesn't matter

$\frac{\partial^2 f}{\partial x_1^2}$, $\frac{\partial^2 f}{\partial x_1 \partial x_2}$, $\frac{\partial^2 f}{\partial x_2^2}$ ← second order

Unordered + with repetition

$f(x_1, x_2, x_3, x_4)$ continuous fifth-order partial derivatives

How many distinct fifth-order partial derivatives does f have, counting derivatives as the same when they differ only in the order of differentiation?

$n = 4$ variables
$k = 5$ derivatives

I) $\frac{\partial^5 f}{\partial x_1^{a_1} \partial x_2^{a_2} \partial x_3^{a_3} \partial x_4^{a_4}}$

$4! = 24$

so this means $\binom{n-1+k}{k} = \binom{8}{4} = \underline{\underline{70}}$

$n = 5$
$k = 4$

or

$\binom{4-1+5}{5} = \binom{8}{5} = \underline{\underline{56}}$

(P2) Poker Deck 52 cards

13 ranks → Kartenwerte

4 suits → Kartensymbole

a hand consists of 5 cards

- A pair two cards of the same rank, 3 diff. ranks
- Three of a kind: 3 cards of the same rank
  2 different
- A straight consists of 5 consecutive cards
- A straight flush all cards of the same suite

10, 9, 8, 7, 6

total hands $\binom{52}{5} = 2\,598\,960$

What is the probability of obtaining two pairs?

i.e. J♥, J♠, 3♣, 3♠, 2♡

one hand 5 cards of 5

→ two hand 10 cards of 52

4 cards – pair

6 cards – different

$\binom{13}{2} = 78$

→ total no of ranks of two pairs × total no of suits in a pair $\binom{4}{2}^2 = 6 \cdot 6 = 36$ (4 suits, 2 per pair)

× cards that aren't the same rank $13 - 2 = 11 \cdot 4 = 44$ (4 suits)

$78 \cdot 36 \cdot 44 = 123\,552 \leftarrow A \quad P(A) = 4{,}75\,\%$

What is the probability of achieving three of a kind?

$\binom{13}{1}$ No of distinct cards · $\binom{4}{3}$ No of distinct suits — first 3 pairs

$\binom{12}{2}$ on the last two pos. no of distinct card numb. · $\binom{4}{1}^2$ for each pos. number of distinct card suit — the rest of hand

$\Rightarrow 2{,}87\,\%$

What is the probability of obtaining a straight

Straight is a sequential following of cards. Not all of them the same suit.

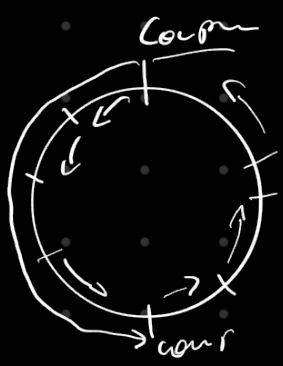
$10 \cdot 4^5 = 10\,240 \approx 0{,}39\,\%$

$\Rightarrow$ ranks above J, Q or K aren't considered numbers but are ranked higher than 10

for consecutive cards the counting begins from A to 10

(P3) A group of 13 friends arrive at the restaurant. To greet one another, everyone shakes hands. How many handshakes are exchanged in total?

$\binom{13}{2} = 78$ ✓



11 People single
2 People couple

not sit aside

→ $(n-1)!$ positions – $2! \cdot 11!$ couple

⇒ $(13-1)! - 2 \cdot 11! = 399\,168000$ ✓

P4 Seven People are playing Secret Santa. Each person draws one name from a hat containing all seven names.

If anyone draws their own name, everyone puts their names back and draws again

→ Variation without repetition? ✓ + inclusion exclusion principle

$\frac{n!}{(n-k)!}$: $n=7!$ $k=7$

↓

5040

↓

7 draws from 7 people

Double checked it, so it's inclusion exclusion too

Howto: define the pieces use the product rule

⟹ What is the chance that nobody gets their own name

No of total out comes: $7! = 5040$

A_i = number of people who draw themselves

$$|\Omega| - |(A_1 \cup \dots \cup A_7)^c|$$

if someone draws their own one $(7-k)!$ possible ordns to get someone

↓

1 person draws it. ← $6! = 720$
2 ppl ← $5! = 120$
3 ppl ← $4! = 12$
4 ppl ← $3! = 6$
5 ppl. ← $2! = 2$

to choose which of these 7 people it is: $\binom{7}{k}$

$$\binom{7}{k} \cdot (7-k)!$$

$$\Rightarrow \sum_{k \geq 1} \binom{7}{k} \cdot (7-k)! \cdot (-1)^k$$

$$\Leftrightarrow |\Omega| - \sum_{k \geq 1} \binom{7}{k} \cdot (7-k)! \cdot (-1)^k$$

R: $5040 - 5040 + 2520 - 840 + 210 - 42 + 7 - 1 = 1854$

(P5) Seven balls are placed into seven cells.
How many of the following arrangements are there?

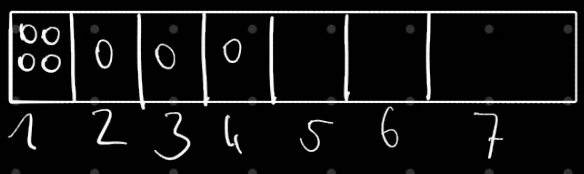
(1) One cell contains 4 balls

(2) two cells contain either two or three balls, all other cells contain neither two nor three balls.

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(1) ○○○○○○○

$n = 7$
$k = 4$



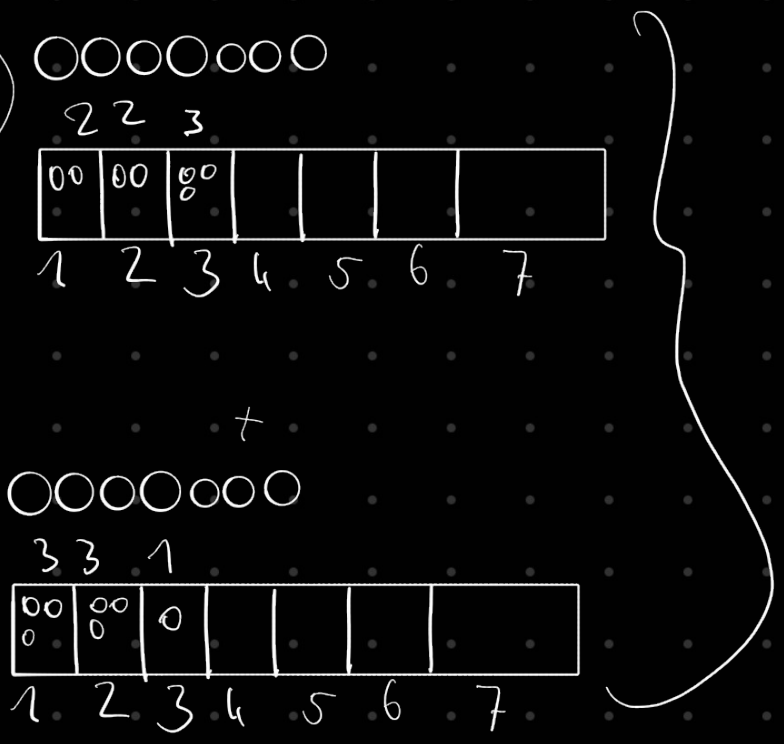
→ One cell contains 4 balls

combination with repetition → $\binom{n-1+k}{k}$

$$\binom{10}{4} = 210$$

Let X_i = the number of cells containing exactly i balls

(2)



Case 1: X_2

→ two boxes hold 2 balls (in total 4 balls)

3 balls remain

5 spaces remain

$\binom{7}{2} = 21$

$+$

$\binom{5}{3} = 10$

→ product rule $21 \cdot 10 = \underline{210}$ ✓

Case 2: $X_{(2,3)}$

6 spaces

5 balls

→ one box hold 2 balls the other one 3 balls

$\binom{7}{2} = 21 \cdot \binom{6}{3} = 120$

$\binom{5}{2} = 60$

product

$21 \cdot 120 \cdot 60 = \underline{151200}$

→ $21 + 120 + 60 = \underline{201}$?

Case 3 X_3

→ two boxes contain 8 balls

$\binom{7}{3} = 35$

$+$

$\binom{5}{1} = 5$ $\quad \underline{\underline{40}}$

sum 1 $210 + 201 + 40 = \underline{451}$

sum 2 $210 + 151200 + 40 = \ddots$

more logical!